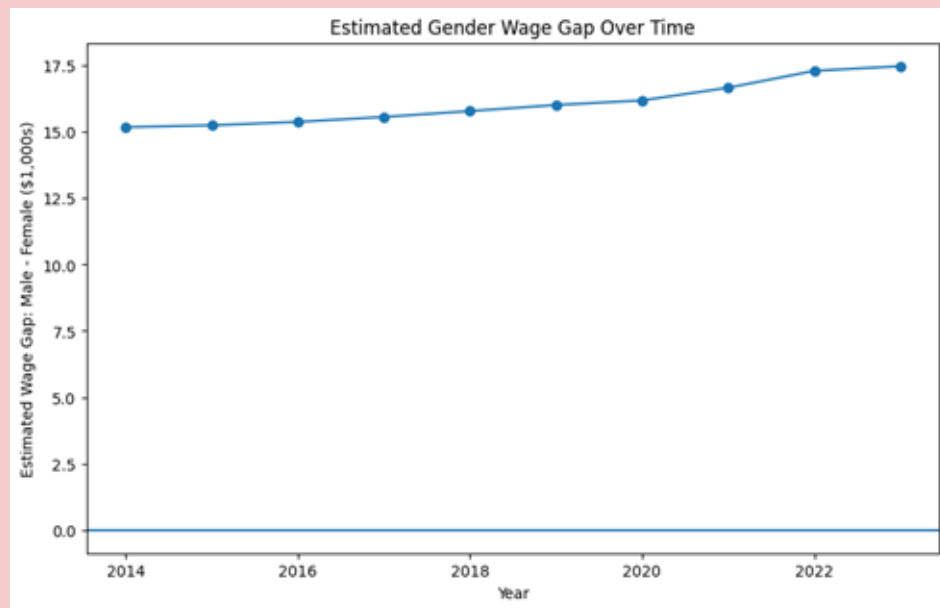


Q: Does the gender wage gap change among dancers & choreographers when controlling for year?

Presented by Shuyi Zhao & Sophia Scott



```
# Model 1: baseline gender wage gap
model1 = smf.wls('wage_midpoint ~ female',
                 data=data,
                 weights=data['Record Count']).
                 fit(cov_type='HC1')

model1.summary().tables[1]
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	61.4110	3.693	16.629	0.000	54.173	68.649
female	-17.4356	4.656	-3.745	0.000	-26.562	-8.309

```
# Model 2: gender wage gap with year controls
model2 = smf.wls('wage_midpoint ~ female + C(Year)',
                 data=data, weights=data['Record Count'])
.model2.summary().tables[1]
```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	54.9682	7.144	7.695	0.000	40.967	68.969
C(Year)[T.2015]	0.3440	9.397	0.037	0.971	-18.074	18.762
C(Year)[T.2016]	1.3885	9.488	0.146	0.884	-17.207	19.984
C(Year)[T.2017]	2.8173	9.595	0.294	0.769	-15.988	21.622
C(Year)[T.2018]	4.4224	9.700	0.456	0.648	-14.589	23.433
C(Year)[T.2019]	6.0846	9.817	0.620	0.535	-13.156	25.325
C(Year)[T.2020]	7.3982	9.915	0.746	0.456	-12.034	26.831
C(Year)[T.2021]	10.0218	10.136	0.989	0.323	-9.845	29.888
C(Year)[T.2022]	14.3136	10.531	1.359	0.174	-6.327	34.954
C(Year)[T.2023]	16.7401	10.737	1.559	0.119	-4.304	37.785
female	-17.4127	4.613	-3.775	0.000	-26.454	-8.372

Model 1: Baseline Gender wage gap

- Estimated Wage Midpoint = $\beta_0 + \beta_1 * \text{Female} + \varepsilon$

Model 2: Baseline Gender wage gap & Residual Plot

- Estimated Wage Midpoint = $\beta_0 + \beta_1 * \text{Female} + \sum \beta_i * \text{Year}_i + \varepsilon$

